

User Acceptance Testing (UAT) Template

Date	
Team ID	
Project Name	AI FAQ Assistant API
Maximum Marks	

Project Overview:

Project Name: AI FAQ Assistant API

Project Description: AI FAQ Assistant API is a NestJS REST backend that lets a business store its FAQs and answer customer questions from them using AI. It finds the right FAQs by meaning, writes a short answer with sources and a confidence score, hands unclear questions to a human, and reports on feedback and unanswered questions. Access is secured with JWT and API keys, with Redis caching, rate limiting and Swagger documentation.

Project Version: v1.0.0

Testing Period: 24 March 2025 to 28 March 2025 (after Sprint-4 ends on 22 March 2025)

Testing Scope:

Features and functionalities to be tested

- Account and access: registration, confirmation email, login, token refresh, roles and API keys
- FAQ management: create, update, delete, Draft / Published / Archived, categories, tags, search, filters, pagination and bulk import
- AI search and answering: embeddings, semantic search, grounded answers with sources, confidence score, “I don’t know” fallback and follow-up questions
- Feedback and escalation: thumbs up / down, escalation to a human agent and the agent queue
- Protection and insight: Redis caching, rate limiting, organisation isolation, handling of LLM failure and analytics
- Developer experience: Swagger documentation

User stories or requirements to be tested

- USN-2 to USN-25 (account and access, FAQ management, AI answering, feedback and escalation, caching, rate limiting, Swagger, agent queue, analytics and API keys)

Out of scope: the customer chat widget or other front-end, large-scale load testing, and developer-only stories USN-1 and USN-26 to USN-30 (setup, automated tests, CI/CD and deployment).

Testing Environment:

URL / Location	Staging: https://<staging-host>/api (Swagger UI: https://<staging-host>/api/docs). Local: http://localhost:3000/api
Stack	NestJS, PostgreSQL with pgvector, Redis, external LLM and embedding API (test key with a low usage limit)
Tools	Swagger UI, Postman, email test inbox (for confirmation and escalation emails)
Test data	Two organisations (A and B), each with about 20 published FAQs on orders, delivery, returns and payments

Credentials (if required):

Role	Username	Password / Key
Admin (Organisation A)	admin@uat.example.com	Provided separately by the project team
Editor (Organisation A)	editor@uat.example.com	Provided separately by the project team
Agent (Organisation A)	agent@uat.example.com	Provided separately by the project team
Editor (Organisation B)	editor.b@uat.example.com	Provided separately by the project team
Client app API keys	One key per organisation	Provided separately by the project team

Test Cases:

Test cases cover both positive and negative scenarios. Record the actual outcome and mark Pass or Fail during testing.

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	Register an organisation account (USN-2, USN-3)	1. Send POST /auth/register with a valid email, password and matching confirm password. 2. Check the response and the inbox.	HTTP 201. Account is created; password is not returned. A confirmation email is received.		
TC-002	Register with an already used email (negative)	1. Register an account. 2. Send POST /auth/register again with the same email.	HTTP 409 Conflict with a clear message. No duplicate account is created.		
TC-003	Register with invalid input (negative)	1. Send POST /auth/register with an invalid email, a weak password, or mismatched confirm password.	HTTP 400 with field-level validation errors. No account is created.		
TC-004	Login with valid credentials (USN-4)	1. Send POST /auth/login with a registered email and the correct password.	HTTP 200. Response contains a JWT access token and refresh token.		
TC-005	Login with wrong password (negative)	1. Send POST /auth/login with a registered email and an incorrect password.	HTTP 401 Unauthorized. No tokens are issued.		
TC-006	Access a protected endpoint without a token (negative)	1. Send POST /faqs without an Authorization header.	HTTP 401 Unauthorized.		
TC-007	Refresh an access token	1. Log in and copy the refresh token. 2. Send POST /auth/refresh with it.	HTTP 200 with a new access token. An invalid or expired refresh token returns 401.		
TC-008	Role restriction: Agent cannot edit FAQs (USN-5, negative)	1. Log in as an Agent. 2. Send POST /faqs with valid FAQ data.	HTTP 403 Forbidden. No FAQ is created.		
TC-009	Editor creates an FAQ (USN-6)	1. Log in as an Editor. 2. Send POST /faqs with question, answer and category.	HTTP 201. FAQ is saved with status Draft and the Editor as author.		

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-010	Editor updates an FAQ (USN-8)	1. Log in as the Editor. 2. Send PATCH /faqs/:id with a changed answer.	HTTP 200. FAQ shows the updated answer.		
TC-011	Organisation isolation (negative)	1. Log in as an Editor from Organisation B. 2. Send GET and PATCH /faqs/:id for an FAQ owned by Organisation A.	HTTP 404 or 403. Organisation A's FAQ is not visible or changed.		
TC-012	Publish an FAQ (USN-9)	1. Send PATCH /faqs/:id/status with Published. 2. Ask a question that matches it. 3. Repeat with a Draft FAQ.	Published FAQ is used in answers. Draft and Archived FAQs are never used.		
TC-013	List, search, filter and paginate FAQs (USN-10, USN-11)	1. Call GET /faqs?page=1&limit=5&search=return&tag=orders&sort=updatedAt.	HTTP 200. Only matching FAQs of the organisation are returned, sorted as requested, with page details.		
TC-014	Delete an FAQ and fetch it again (USN-8)	1. Send DELETE /faqs/:id. 2. Send GET /faqs/:id. 3. Ask a question that the FAQ used to answer.	Delete returns 200/204. The FAQ returns 404 and is no longer used in answers.		
TC-015	Bulk import FAQs (USN-15)	1. Send POST /faqs/import with a CSV that has valid rows and two invalid rows.	Valid rows are imported as Draft FAQs. Invalid rows are listed with line numbers and reasons.		
TC-016	Embedding created for a new FAQ (USN-13)	1. Create and publish an FAQ. 2. Check that it can be found by a semantic search.	The FAQ is searchable within a short time. An embedding failure is logged and retried.		
TC-017	Semantic search finds differently worded questions (USN-14)	1. Publish an FAQ titled "How do I return an item?". 2. Call POST /search with "can I send my order back".	The FAQ appears at or near the top with a similarity score.		
TC-018	Ask a question and get a grounded answer (USN-16)	1. Send POST /ask with a question covered by a published FAQ.	HTTP 200. A short answer is returned with source FAQ IDs and a confidence score. The answer has no facts outside those FAQs.		

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-019	Question with no matching FAQ (USN-17, negative)	1. Send POST /ask with a question unrelated to any FAQ.	HTTP 200 with a clear "I don't know" reply, low confidence and an escalation option. No answer is invented.		
TC-020	Ask with an empty question (negative)	1. Send POST /ask with an empty or missing question.	HTTP 400 with a validation error. The LLM is not called.		
TC-021	Follow-up question in the same session (USN-18)	1. Ask "What is your return policy?". 2. In the same session ask "How long do I have?".	The second answer uses the first question as context. A new session ID starts without that context.		
TC-022	Thumbs up or down feedback (USN-19)	1. Send POST /feedback with an answer ID and a thumbs-down. 2. Send a second vote for the same answer in the same session.	First vote is saved. The duplicate vote is rejected or replaces the first.		
TC-023	Repeated question is served from cache (USN-20)	1. Send the same POST /ask twice and compare time and logs. 2. Edit the matching FAQ and ask again.	The second answer is faster and no second LLM call is logged. After the FAQ edit, a fresh answer is generated.		
TC-024	Rate limit exceeded (USN-21, negative)	1. Send requests above the configured limit for one API key.	HTTP 429 with retry information. Access resumes after the window.		
TC-025	LLM provider unavailable (USN-12, negative)	1. Use an invalid LLM key or block the provider URL. 2. Send POST /ask.	A graceful 502/503 message is returned. FAQ management still works and the API stays up.		
TC-026	Escalated question reaches an agent (USN-23)	1. Ask an unanswerable question and choose to contact a human. 2. Log in as an Agent and call GET /escalations. 3. Mark the item resolved.	The escalation appears with its context and an email alert is sent. After resolving, its status is Resolved.		

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-027	Analytics summary (USN-24)	1. Ask several questions and give feedback. 2. Call GET /analytics/summary?from=...&to=... as an Admin.	Top questions, unanswered rate and satisfaction match the activity. A non-admin gets 403.		
TC-028	Access with an API key (USN-25)	1. Send POST /ask with a valid API key header. 2. Repeat with an invalid or revoked key.	A valid key returns 200 using only its organisation's FAQs. An invalid or revoked key returns 401.		
TC-029	Swagger documentation is available (USN-22)	1. Open /api/docs in a browser. 2. Click Authorize and enter a valid JWT. 3. Run one endpoint from the page.	All endpoints are listed with request and response schemas. The test call works with the token.		

Bug Tracking:

Severity: Low / Medium / High. Status: Open / In Progress / Closed. Add rows as needed.

Bug ID	Bug Description	Steps to reproduce	Severity	Status	Additional feedback
BG-001					
BG-002					
BG-003					
BG-004					
BG-005					

Sign-off:

Role	Name	Date	Signature
Tester			
Project Manager			
Product Owner			

Notes:

- Ensure that all test cases cover both positive and negative scenarios.
- Encourage testers to provide detailed feedback, including any suggestions for improvement.
- Bug tracking should include details such as severity, status, and steps to reproduce.
- Obtain sign-off from both the project manager and product owner before proceeding with deployment.
- Use test organisations and a low-limit LLM API key in UAT so that testing does not affect real customers or incur high AI costs.